

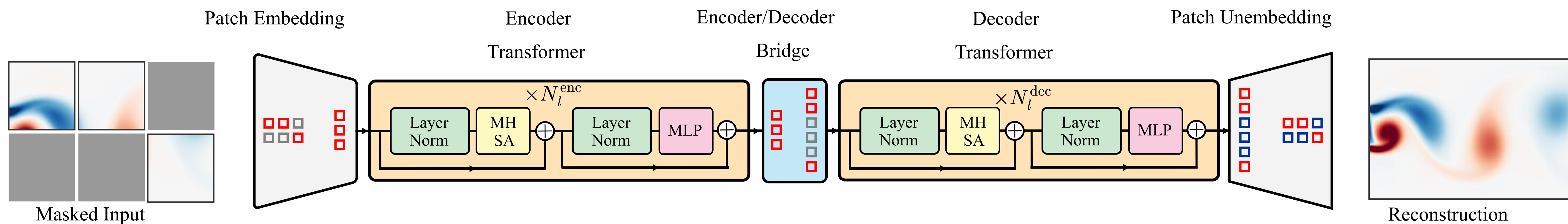
Vision Transformer Masked Autoencoders for Flow Reconstruction

Ben Eze, Luca Magri, Andrea Nóvoa

Aims:

- Reconstruct 2D flow-fields from masked PIV-like data with low error
- Achieve robustness to 30dB noise¹

ViT-MAE Architecture



- Transformer architecture² kept as close to LLM implementation, using **flattened image patches** instead of tokens³
- Encoder acts on **unmasked** patches only, with positional information given by positional encoding
- Decoder transformer mirrors encoder structure, acting on **all** patches
- U-Net layer attenuates patching artefacts

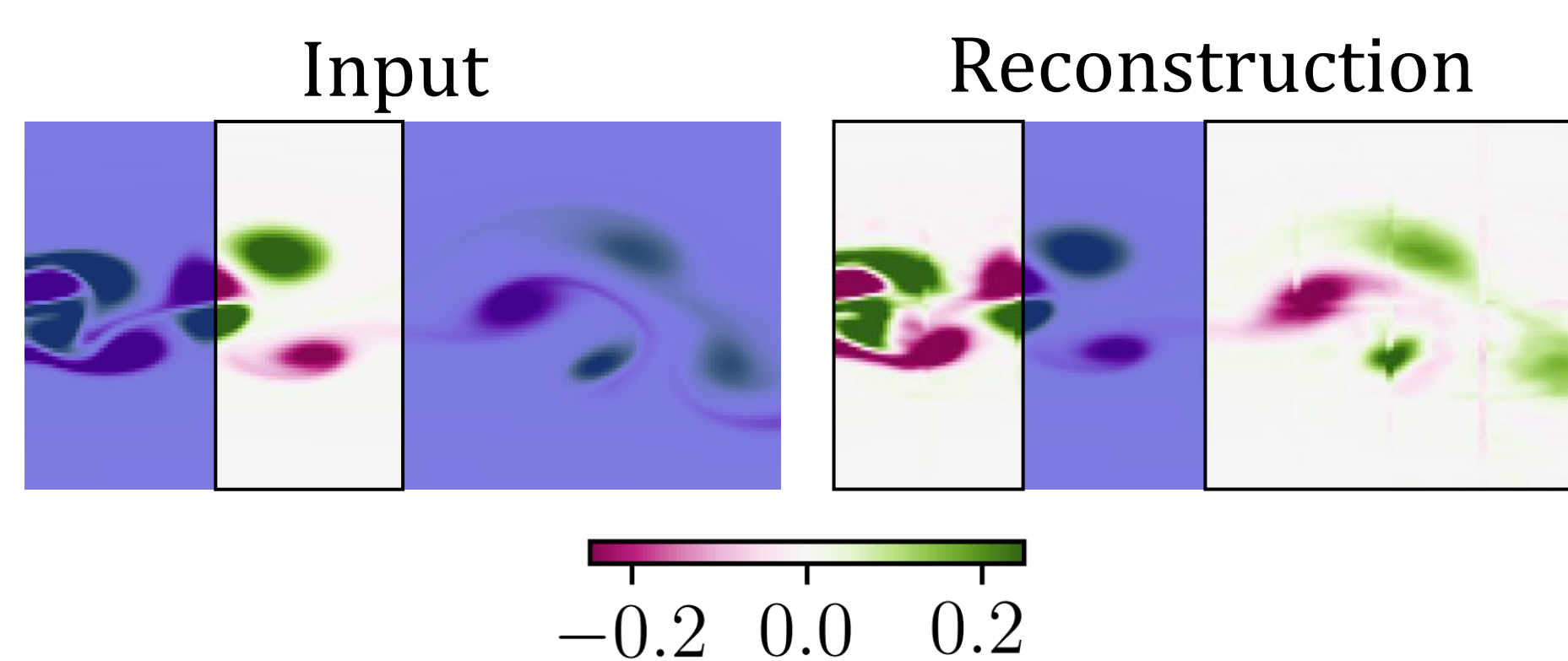
Baseline Reconstruction

Data

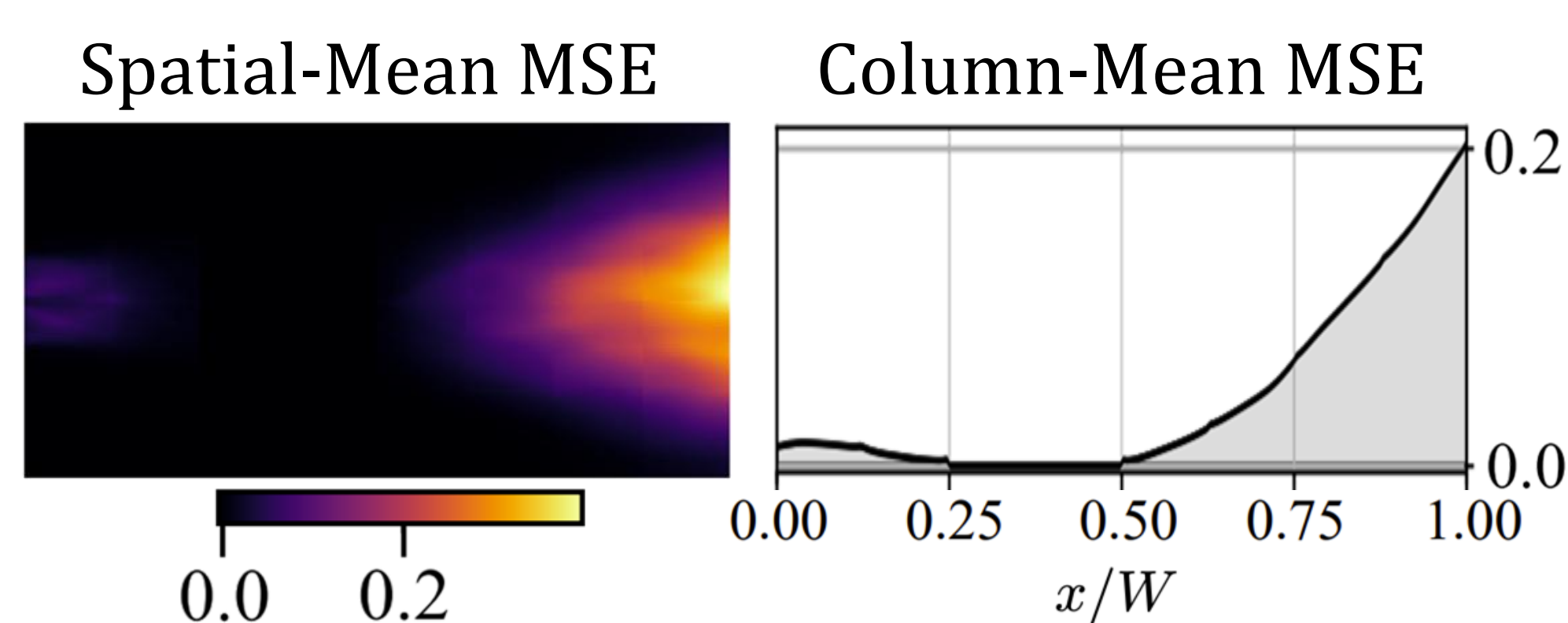
- Dataset contains snapshots of chaotic flow behind two adjacent cylinders

Example reconstruction

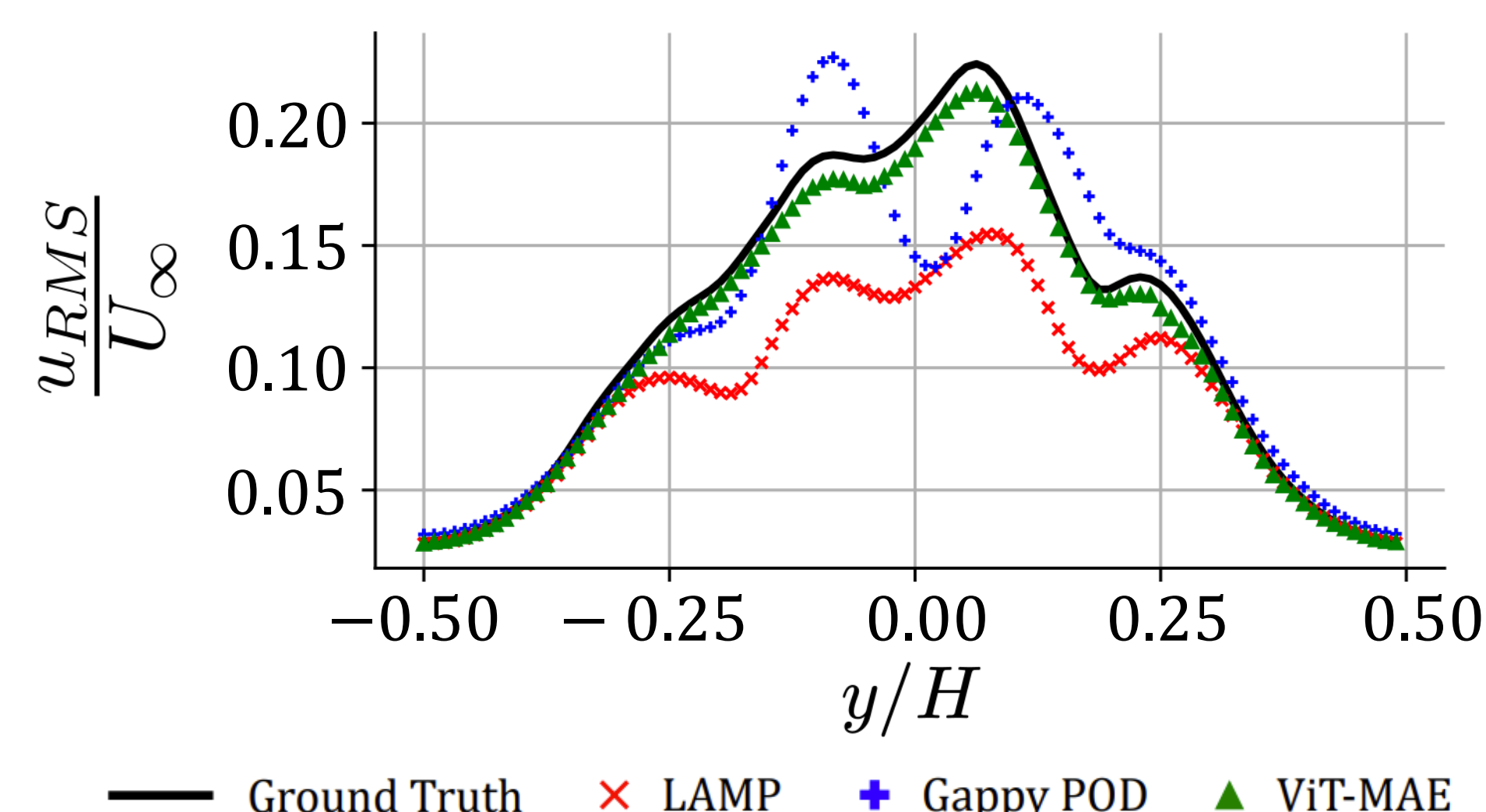
- Vorticity field of input data (ω_z) and reconstructed flow (ω_z^*)



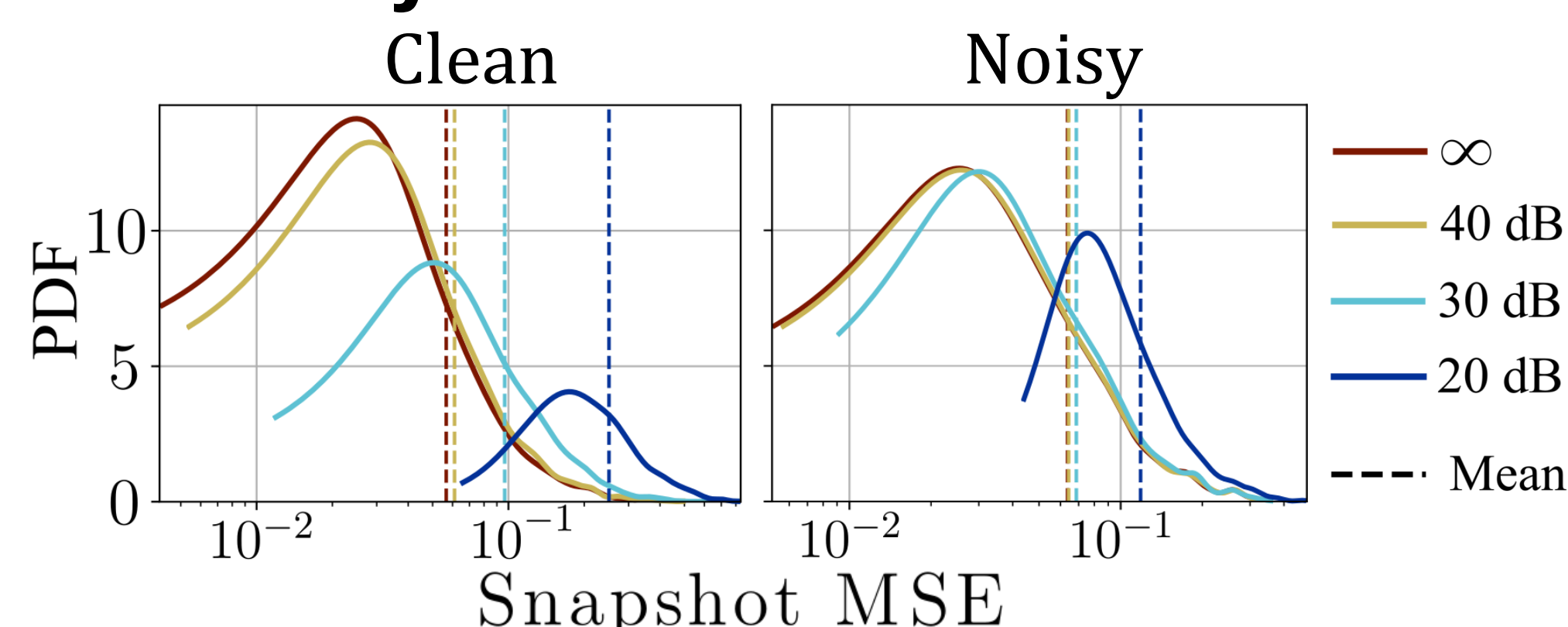
Mean reconstruction error (u, v) increases in the wake



ViT-MAE outperforms Gappy POD & LAMP⁴ in reconstructing velocity profiles



Noise robustness improves when trained with noisy data

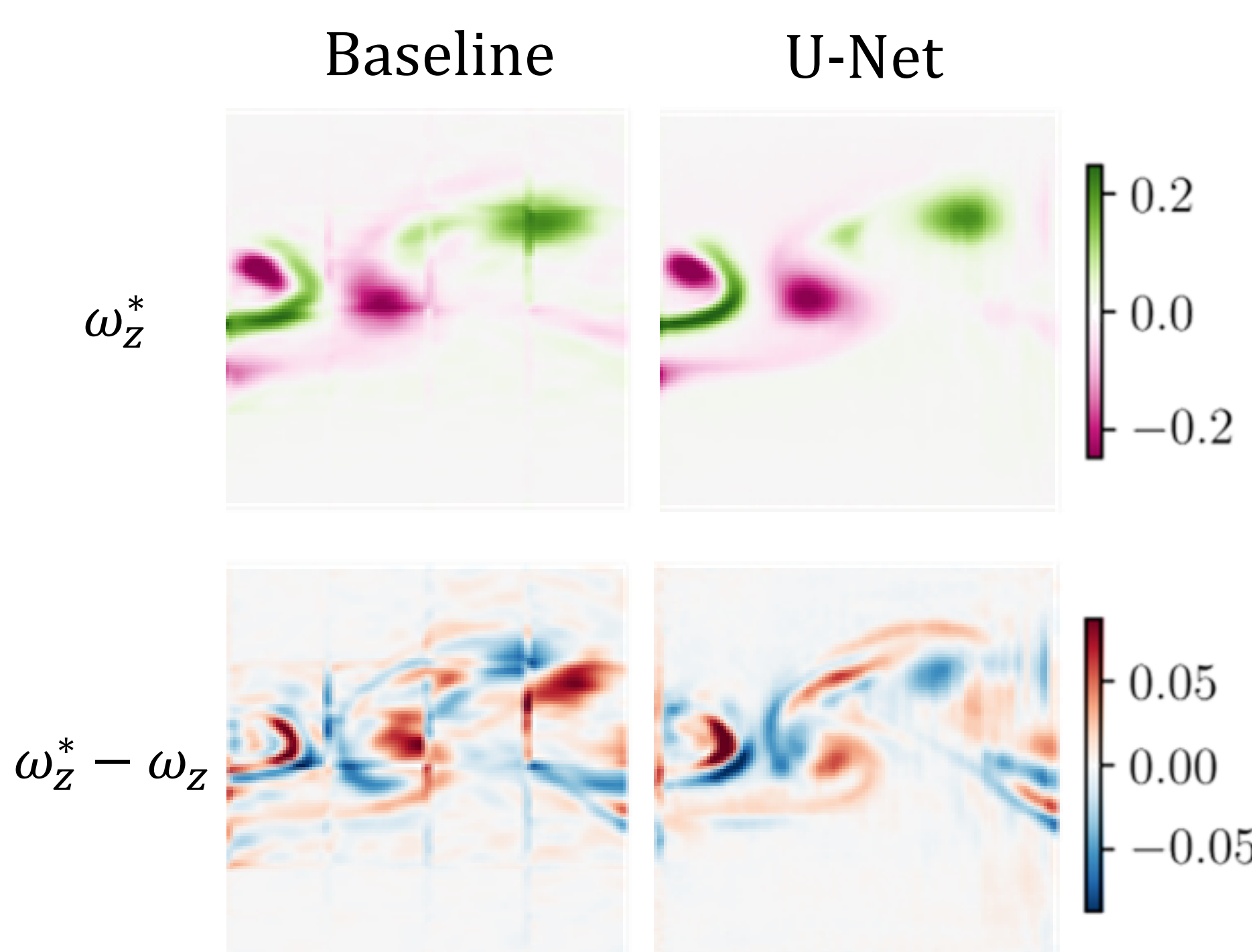


Artefact Removal

- U-Net acts on output of patch unembedding (i.e. in **physical space**)
- An additional loss term penalises error in the vorticity

$$\mathcal{L}_{\omega_z}^{\text{pred}} = \|\tilde{\mathbf{X}}^* - \mathbf{X}^{\text{test}}\|_2^2 + \beta_{\omega} \|\omega_z(\tilde{\mathbf{X}}^*) - \omega_z(\mathbf{X}^{\text{test}})\|_2^2$$

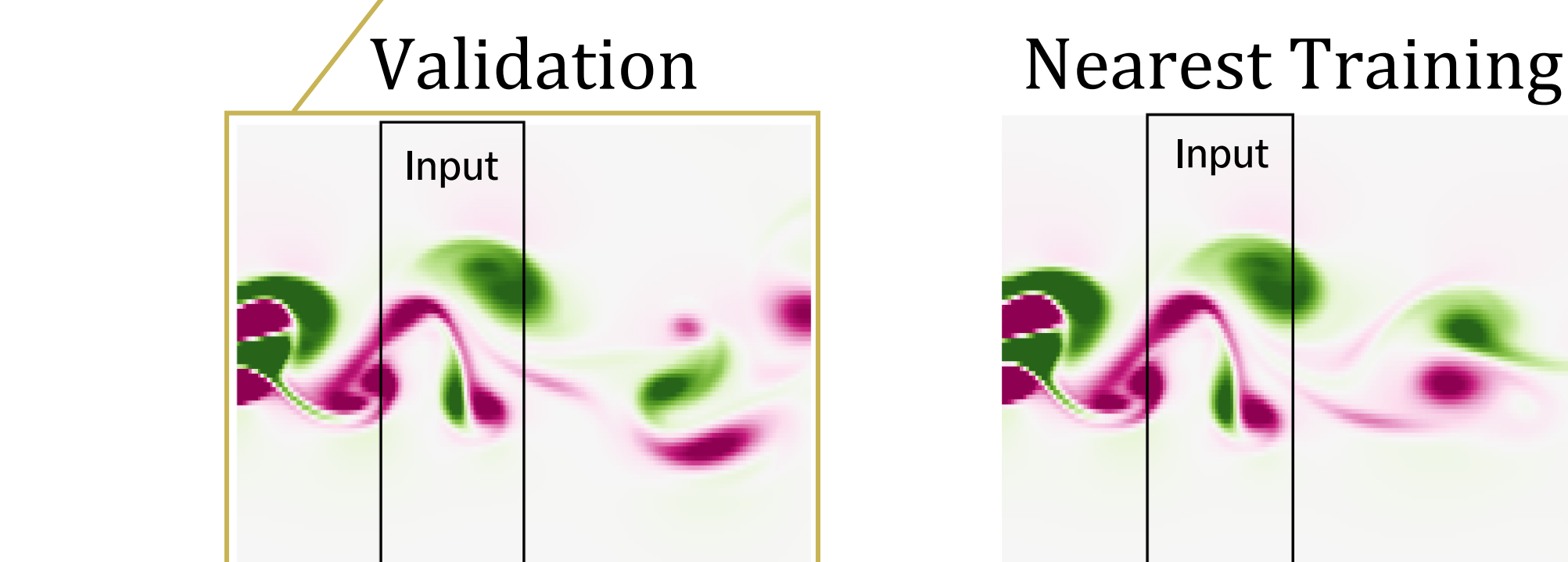
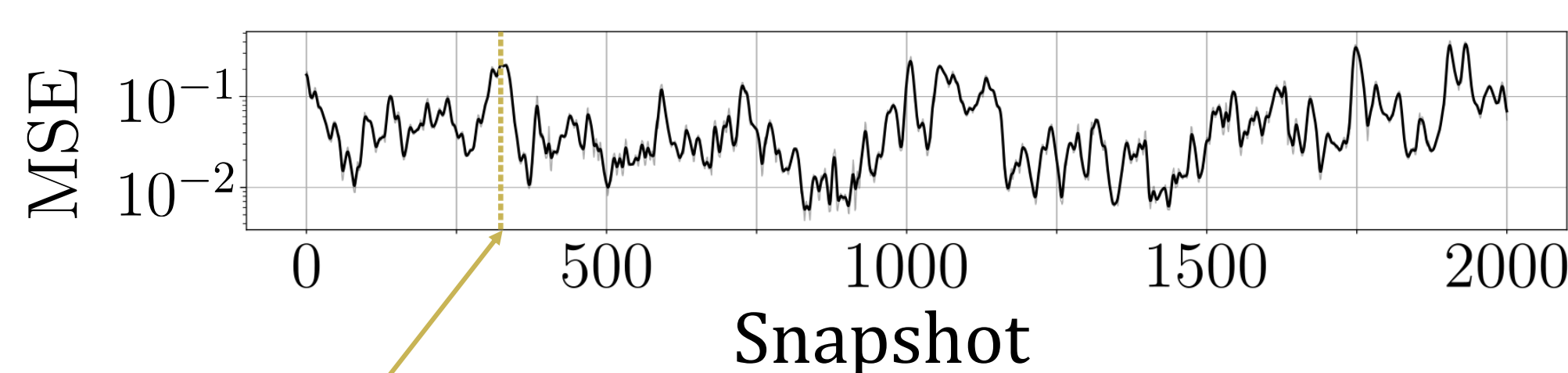
U-Net removes artefacts from patching



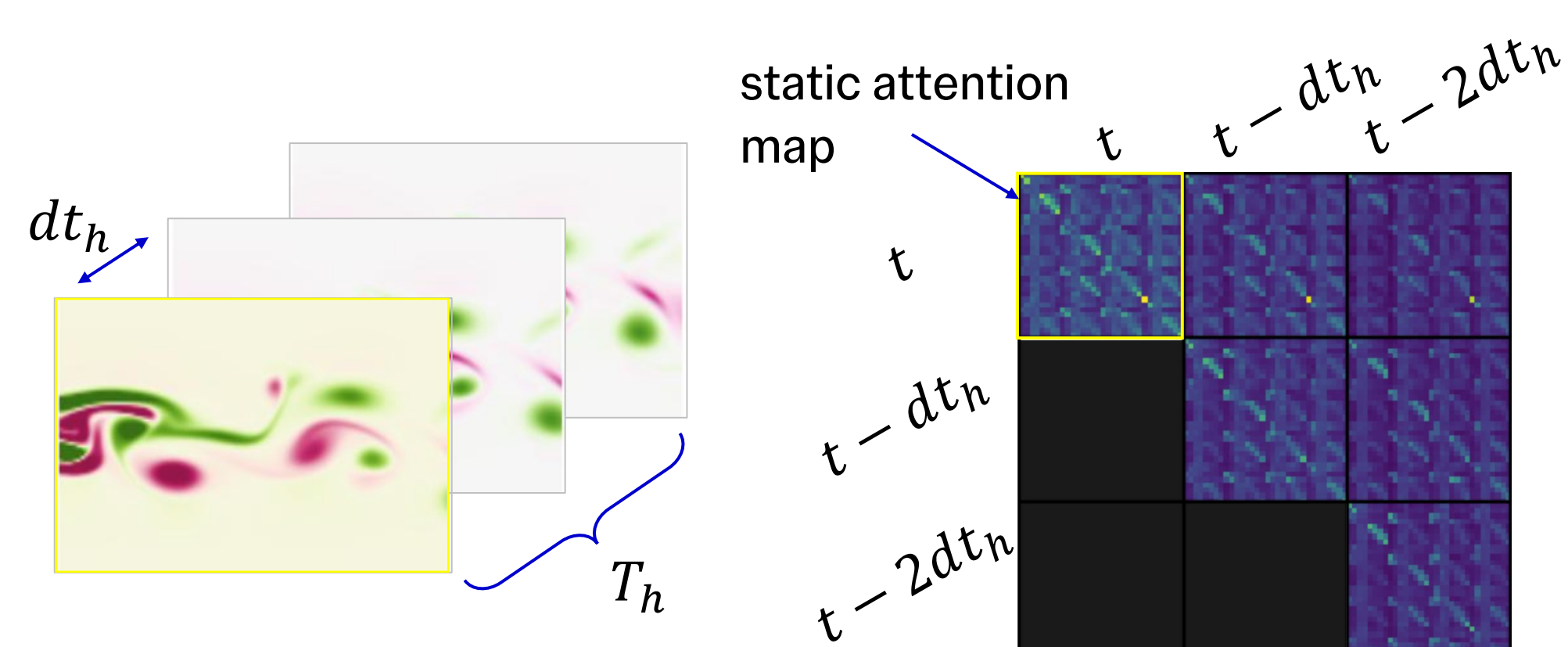
- Best results when **trained in conjunction** with the rest of the model

Time History

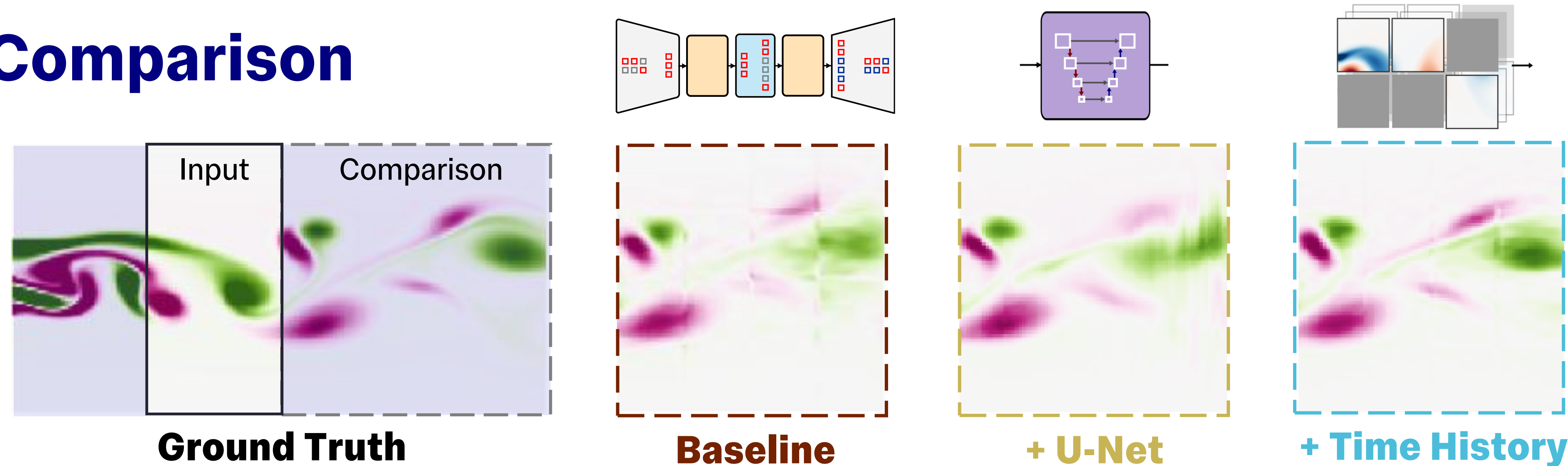
- Error peaks often occur when similar input flows in the training data correspond with different states downstream



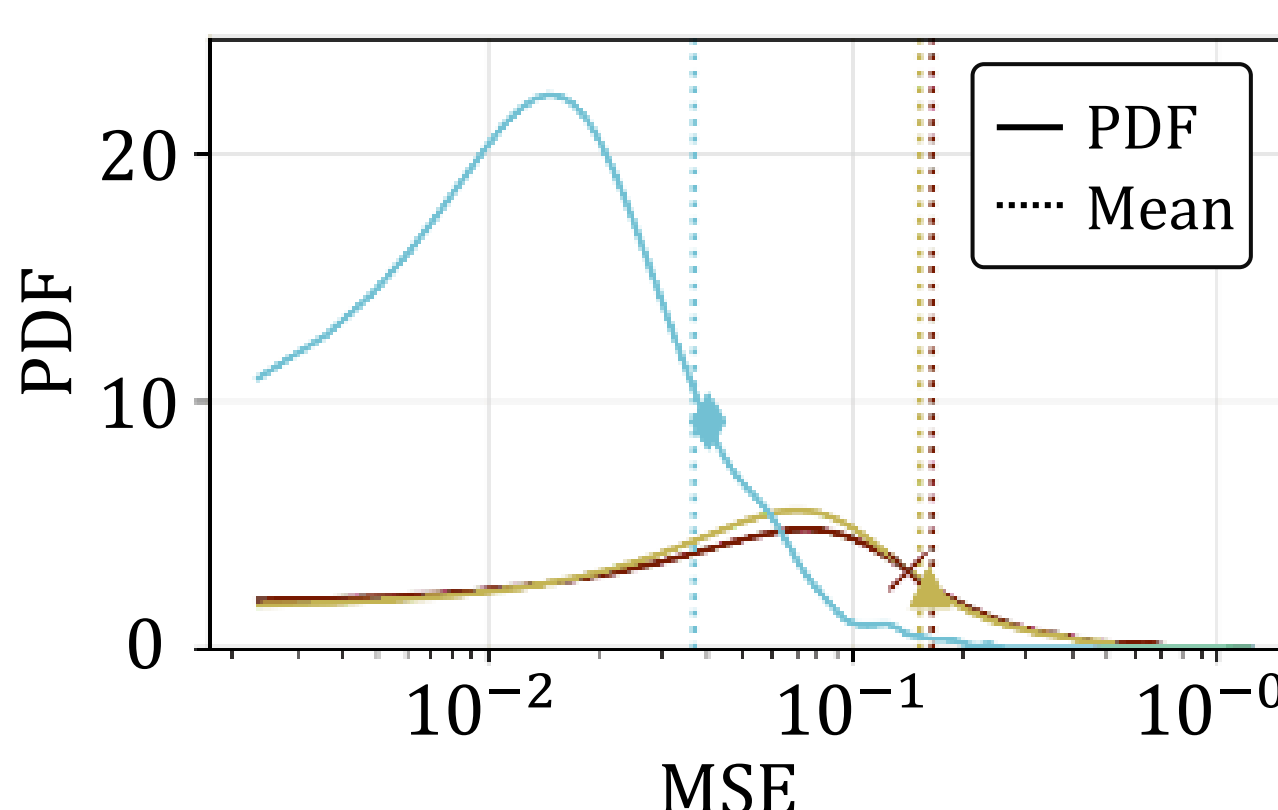
- The attention mechanism is expanded to process a history of up to T_h input samples



Comparison



- U-net improves the smoothness of the reconstructions, but only minimally lowers the overall MSE
- Adding time history significantly reduces the MSE, better capturing the flow structures



References

- Steven J. Beresh. *Evaluation of PIV Uncertainties using Multiple Configurations and Processing Techniques*
- Ashish Vaswani et al. *Attention is all you need*
- Alexey Dosovitskiy. *An Image Is Worth 16 x 16 Words: Transformers for Image Recognition at Scale*
- Ben Eze, Luca Magri, and Andrea Nóvoa. *Latent Attention on Masked Patches for Flow Reconstruction*.

